

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0

- **Coursework:** Data Structures & Algorithms, OOP, Computer Architecture, Linear Algebra, Discrete Math

EXPERIENCE

MindMosaic

Software Engineering Intern

Aug 2025 – May 2026

East Lansing, MI

- Built and shipped an interactive knowledge-graph platform that keeps dense, interconnected content navigable as it grows, delivering a production MVP to real users on a polyglot Java Spring Boot, Neo4j, and PostgreSQL stack.
- Migrated hot read paths off recursive SQL self-joins onto a Neo4j graph layer so relationships stay first-class, cutting query latency 48% (850ms to 440ms) and making multi-hop traversals 3x faster than the equivalent SQL.
- Designed a dual-database layer that keeps transactional records in PostgreSQL while serving relationship-heavy reads from Neo4j, routing each query class to the store that answers it fastest instead of forcing one engine to do both jobs.
- Owned the Spring Boot backend serving the platform's core CRUD and graph-traversal APIs, structuring endpoints around the dual-store model so the client never has to know which database backs a given read.
- Authored an AWS deployment script and managed the environment as code with Terraform, replacing a manual 30-minute multi-service bring-up with a single-command, version-controlled provision.

Claude Builder Club @ MSU

Co-Founder & Vice President

Oct 2025 – Present

East Lansing, MI

- Co-founded and grew an MSU engineering community from zero to 150+ members, replacing a sprawl of spreadsheets, Slack threads, and manual invites with a single club-operations platform I designed and built.
- Built the club's mentorship lineage as an interactive 3D radial graph in react-three-fiber, hand-writing a proportional-arc layout that prevents edge crossings and a custom camera that autofits each family to the viewport.
- Automated member onboarding so an accepted applicant is provisioned into the right GitHub teams, Slack channels, Discord roles, and Google Drive access in a single flow, cutting onboarding from 15 minutes to seconds.
- Built event check-in as a single atomic database RPC that validates a QR token, records attendance, and awards points server-side, making door-time check-ins concurrency-safe and idempotent, sustaining 150+ check-ins per event.

PROJECTS

Dadei | *React, FastAPI, Redis*

dadei.app

Ambient voice assistant that turns overheard conversation into human-approved calendar, email, and task actions

- Built a React client that streams microphone audio up over a WebSocket with client-side Web Audio processing, receiving incremental live captions back in under 300ms as the backend transcribes each utterance.
- Decoupled background workers from the live FastAPI service behind a Redis Streams event bus with consumer groups, dead-letter routing after 5 retries, and stale-claim recovery, giving at-least-once delivery that pub/sub dropped.
- Solved the same-room, multi-mic problem with two arbitration layers, a live utterance-owner election and a 400ms windowed quality election on SNR, clarity, and cadence, cutting duplicate transcription by 50%.

fliks | *Go, PostgreSQL*

github.com/fliks-gg

Multi-game platform that crowd-validates gameplay skill and mints shareable certification cards

- Architected an ML rating service that grades each clip independently and shows its verdict beside the crowd's, with those crowdsourced ratings serving as the labeled data the classifier trains on.
- Made ratings immutable and gated commenting behind a submitted rating, so the certification signal cannot be retroactively gamed and social participation feeds skill data instead of diluting it.

TECHNICAL SKILLS

Languages JavaScript/TypeScript, Python, Java, SQL, Go
Frameworks React, FastAPI, Spring Boot
Databases PostgreSQL, Redis, Neo4j